

Multiple Choice (10 marks)

1. Cholesterol and fatty acids absorbed from the intestinal lumen are packaged into which types of lipoproteins and secreted from the enterocytes into the lymph?
 - (a) Chylomicrons only
 - (b) LDL only
 - (c) VLDL and LDL
 - (d) Chylomicrons and HDL
 - (e) VLDL and HDL
2. A burial site where the body is allowed to decompose naturally without a casket is called a _____ cemetery
 - (a) Eco-friendly
 - (b) Sunshine
 - (c) Forest
 - (d) Rapid decay
 - (e) Organic
3. Older adults are able to improve their memories and reduce their anxiety about declining memory when
 - (a) They engage in social activities and interact with peers
 - (b) They practice mindfulness and meditation to reduce stress levels
 - (c) They engage in physical activity regularly
 - (d) They regularly solve puzzles and engage in brain-stimulating activities
 - (e) They learn that many aspects of memory do not decline and some even get better
4. The energy charge of the cell is:
 - (a) the extent to which the total adenine nucleotide pool is phosphorylated.
 - (b) the difference between the total energy input and output of the cell.
 - (c) generated by the sodium-potassium ATPase.
 - (d) the amount of ATP available in the cell.
 - (e) the energy required for the maintenance of the cell's structure.
5. The following amino acids are all essential components of the diet:
 - (a) leucine, methionine and threonine

- (b) leucine, cysteine and glycine
- (c) proline, tyrosine and serine
- (d) valine, glycine and arginine
- (e) methionine, serine and tyrosine

True / False (10 marks)

Answer True or False

6. True or False: Nearly one-third of grieving individuals appear to require some form of professional help during their mourning process.
7. True or False: A person weighing 62 kg with a drug dose of 15 mg/kg requires a total dose of 18.6 grams.
8. True or False: Type IIb muscle fibers are characterized by low fatigue resistance compared to other muscle fiber types.
9. True or False: Meat and dairy products are consumed in lower quantities in Wales and Scotland compared to fruits and vegetables.
10. True or False: A normal resting heart rate for an adult is considered to be between 70 and 90 beats per minute.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between optimism in older adults and longevity, analyzing how pessimistic outlooks may influence health outcomes and lifespan.
12. Discuss the energy expenditure of various organs in the human body, focusing on why the intestines have the lowest energy expenditure per unit of body mass.

End of Paper